Methodology 1 — Live Demo Guide

Project: DoS Attack Analysis in SDN-Based VANETs

Purpose: A 10-minute practical demonstration showing SUMO mobility, Mininet-WiFi VANET emulation, Ryu SDN control, controlled DoS traffic, OpenFlow filtering, and preservation of legitimate traffic.

1. Demo Story — What to Explain

Explain the system in this order: SUMO generates vehicle mobility; Mininet-WiFi represents the wireless VANET; AP1 and AP2 connect vehicles to the SDN switch; Ryu acts as the controller; sta3 is the controlled attacker node; and the OpenFlow rule blocks UDP traffic from the configured attacker source. All attack traffic is confined to the local Mininet-WiFi environment.

2. Start SUMO-GUI

Open a terminal and run:

```
cd ~/sdn_vanet_sumo
sumo-gui -n network.net.xml -r routes.rou.xml --start --delay 200
```

Show: the road network and moving vehicles.

Say: “SUMO is used to generate the vehicle mobility scenario. The vehicles move according to the defined road network and traffic flow.”

Close SUMO-GUI after briefly showing the mobility.

3. Start Ryu Controller

Open another terminal and run:

```
ryu-manager ~/dos_defense.py
```

Show: the Ryu log and the message that the DoS defense was installed.

Say: “Ryu is the SDN controller. It installs an OpenFlow filtering rule to block UDP traffic originating from the configured attacker node.”

4. Start Mininet-WiFi

Open a third terminal and run:

```
sudo mn -c
sudo python3 ~/sdn_vanet_2ap.py
```

Wait for the **mininet-wifi>** prompt.

Topology: sta1 and sta2 are legitimate stations; sta3 is the controlled attacker; AP1/AP2 provide wireless access; s1 is the SDN switch; c0 is the Ryu controller.

5. Prove the SDN Devices Are Connected

At mininet-wifi> run:

```
sh sudo ovs-vsctl show
```

Show: AP1, AP2 and s1 with Controller tcp:127.0.0.1:6653 and is_connected: true.

Say: “The Open vSwitch devices are connected to the Ryu controller through OpenFlow.”

6. Show the Defense Rule

Run:

```
sh sudo ovs-ofctl -O OpenFlow13 dump-flows ap2
```

```
Find the rule:

priority=100,udp,nw_src=10.0.0.3 actions=drop
```

Say: “This OpenFlow rule drops IPv4 UDP packets whose source address is 10.0.0.3. The attacker is associated with AP2, so AP2 is the first filtering point.”

7. Show Normal Communication

```
Run:

sta1 ping -c 5 10.0.0.3
```

Say: “Before demonstrating the attack, I verify that communication is functioning.”

Recorded baseline for the formal experiment: 20/20 packets received, 100% PDR, average latency 0.792 ms.

8. Demonstrate the Controlled DoS Attack

```
Start the UDP receiver:

sta1 iperf -s -u &

Then generate the controlled UDP traffic from the attacker:

sta3 iperf -c 10.0.0.1 -u -b 10M -t 20
```

Important: The traffic is generated only between Mininet-WiFi namespaces. Do not direct it at an external system.

Say: “This is a controlled 10 Mbps UDP DoS traffic experiment inside the emulator.”

A warning about the final UDP ACK may appear because the traffic is being filtered. This is expected in this defense demonstration.

9. Prove That the Defense Dropped the Attack

```
Run:

sh sudo ovs-ofctl -O OpenFlow13 dump-flows ap2
```

The n_packets value for the priority-100 UDP drop rule should have increased substantially.

Formal experiment result: **18,036 attacker packets dropped** and **27,270,432 bytes dropped** (about 26.01 MB). The live counter may be different because of timing; the important point is that it increases during the attack.

Say: “The increase in the OpenFlow counter confirms that the defense rule actively filtered the attacker packets.”

10. Prove Legitimate Traffic Still Works

```
Run:

sta1 iperf -s -u &
sta2 iperf -c 10.0.0.1 -u -b 2M -t 20
```

Formal result: **2.10 Mbps** legitimate UDP throughput with the defense active, matching the normal legitimate UDP measurement of 2.10 Mbps.

Say: “The defense is source-specific, so the configured attack traffic is blocked while the tested legitimate traffic continues.”

11. Show the Final PDF Results

Result	Measured value
Normal ICMP PDR	100%
Normal ICMP average latency	0.792 ms
DoS ICMP PDR	100%

DoS average latency	68.305 ms
Latency increase	67.513 ms
Latency increase	8524.4%
DoS UDP throughput	10.5 Mbps
Defense packets dropped	18,036
Defense bytes dropped	27,270,432 bytes
Approx. filtered data	26.01 MB
Legitimate UDP with defense	2.10 Mbps

Key discussion: In this experiment, the DoS attack did not reduce ICMP PDR; instead, it caused severe latency degradation. Average latency increased from 0.792 ms to 68.305 ms, approximately 86.24 times the baseline. The SDN filtering rule then dropped attacker UDP traffic while the tested legitimate UDP flow continued at 2.10 Mbps.

12. Questions Your Mam May Ask

Why use SUMO?

SUMO provides the vehicle mobility scenario and produces realistic vehicle movement data for the VANET experiment.

Why Mininet-WiFi?

It provides the wireless network emulation used to represent the VANET communication environment.

Why Ryu?

Ryu is the SDN controller used to manage the OpenFlow switches and install the defense rule.

What is the attacker?

sta3 is the controlled attacker station inside the local Mininet-WiFi environment.

What does the defense do?

It statically filters IPv4 UDP traffic from source 10.0.0.3 using an OpenFlow drop rule.

Is this rate limiting?

No. The current implementation is source-based UDP filtering. A true rate limiter would require an OpenFlow meter or another explicit rate-control mechanism.

Why did PDR remain 100% during DoS?

For this particular topology and load, the main observed impact was latency degradation rather than ICMP packet loss.

What is the strongest result?

Average latency increased from 0.792 ms to 68.305 ms during the attack, while the defense recorded 18,036 dropped attacker packets.

Does the defense block all traffic?

No. The rule specifically matches UDP traffic from the configured attacker source. The tested legitimate traffic from another station continued.

What is Methodology 2?

It will extend the work toward mobility-aware DoS detection using vehicle mobility characteristics. It is not part of this Methodology 1 demo.

13. Emergency Recovery Commands

If something goes wrong during the demo, do not panic. Exit Mininet with:

```
exit
```

```
Then clean the network:
```

```
sudo mn -c
```

Restart Ryu and Mininet-WiFi from the beginning.

14. One-Sentence Project Summary

“In Methodology 1, I used SUMO for vehicle mobility, Mininet-WiFi and Open vSwitch for the SDN-based VANET environment, Ryu for centralized control, a controlled UDP DoS source for attack injection, and an OpenFlow source-based filtering rule to mitigate the attack while preserving tested legitimate communication.”